# Here are a few of the experiments hitching a ride on the Air Force’s secret space plane

By **Nathan Strout**
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The Air Force’s X-37B Orbital Test Vehicle will return to orbit May 16. (Jeremy Webster/ U.S. Air Force)

When the secretive X-37B space plane returns to orbit on May 16, it will be carrying more experiments than it has on any previous mission, **including one that will transmit solar energy from space to the ground via microwave energy.**

"The X-37B team continues to exemplify the kind of lean, agile and forward-leaning technology development we need as a nation in the space domain," said U.S. Space Force Chief of Space Operations Gen. John "Jay" Raymond. "Each launch represents a significant milestone and advancement in terms of how we build, test, and deploy space capabilities in a rapid and responsive manner."

The unmanned X-37B, which returned from its last and longest flight in October, is scheduled to launch May 16 from Cape Canaveral Air Force Station, Florida. While an earlier Space Force launch of a GPS III satellite was delayed due to the COVID-19 situation, the X-37B launch has remained on track. Despite being launched by the Space Force, the X-37B remains an Air Force platform.

The military has been elusive about what the Boeing-built space plane has been doing on its various missions, beyond noting that it has been used for a number of on orbit experiments. The vehicle has spent a cumulative 2,865 days on orbit, with its last flight being the longest at a record breaking 780 days.

In a May 6 press release, the Space Force opened up about some of the experiments that would hitch a ride into orbit aboard the X-37B, most notably one that will deliver solar power to the ground from space via radio frequency microwave energy.

That experiment is likely related to the Air Force Research Laboratory's Space Solar Power Incremental Demonstrations and Research (SSPIDR), **an effort to collect solar energy with high-efficiency solar cells, convert it to radio frequency, and then beam it to earth.** That technology could provide an uninterrupted energy source to expeditionary forces at forward operating bases that have limited access to traditional power sources.

"The Space Solar Power Incremental Demonstrations and Research (SSPIDR) Project is a very interesting concept that will enable us to capture solar energy in space and precisely beam it to where it is needed," Col. Eric Felt, director of AFRL's Space Vehicles Directorate, said in an October statement on the effort. “SSPIDR is part of AFRL’s ‘big idea pipeline’ to ensure we continue to develop game-changing technologies for our Air Force, DoD, nation, and world.”

AFRL has awarded Northrop Grumman a $100 million contract to support space-based experiments supporting SSPIDR.

The X-37B will also deploy the FalconSat-8, an educational small satellite developed by the U.S. Air Force Academy that will carry five experimental payloads. Also on board will be two National Aeronautics and Space Administration experiments that will study the effects of radiation and the space environment on seeds used for food products.

One reason the vehicle will carry more experiments than prior mission is the attachment of a new service module to the aft of the spacecraft, which will host multiple experiments.

“This launch is a prime example of integrated operations between the Air Force, Space Force, and government-industry partnerships,” said Air Force Chief of Staff Gen. David Goldfein. “The X-37B continues to break barriers in advancing reusable space vehicle technologies and is a significant investment in advancing future space capabilities.”


## About **Nathan Strout**

Nathan Strout covers space, unmanned and intelligence systems for C4ISRNET.